Collaborative Docs >

Discussion Agenda 04/26/2013

Below is a discussion agenda in an approximate order of importance for the second Web Conference of the Project AGORA (Apr. 26, 9am PDT/6pm CEST). Members of the AGORA Steering Committee (Abel, Gnedin, Madau, Mayer, Primack, Teyssier, Wadsley) discussed some of the items below on Apr. 2.

Contents

PLEASE READ THE IMPORTANT ACTION ITEMS IN RED BOLDFACE !

1. Progress Report (as of 04/26/2013)

The following table summarizes the progress made so far in each Working Group.

	WORKING GROUP	PROGRESS
I	Common Physics and Introduction to Project	*[1] Flagship Paper* - The draft of the AGORA Flagship paper is being written thanks to the contributions from many people, and is at the following location. The paper is being prepared to be submitted to ApJS. Not for public consumption yet. http://[link_outdated] - We thank all the people who have contributed to the writing of this article. Please note that the author list of this paper is not final, and at the moment *purely* alphabetical. The rest of the decision is left to the Steering Committee so they can later decide the authorship and, if needed, its order (as discussed **before**; see the footnote at the bottom of the first page of the linked paper). If you would like to learn about the authorship policy for the Flagship paper and all other subsequent papers of the AGORA Project as agreed in the 1st Web Conference, see **this page**. Please also note that your contributions had to be modified to keep the consistency of the article. Red colored comments are by the Steering Committee. *[2] Common Physics Package Introduced* - Please see Sections 3.1 to 3.7 of the paper linked above for detailed information. *[3] DM-only Proof-of-Concept Flagship Runs for Flagship Paper* - Please see Sections 5.1 to 5.3 of the paper linked above for detailed information. - In the Flagship comparison of a ~1e11 Ms halo, we have so far embraced a philosophy of letting each group to take as much time as needed to figure out how to read in the MUSIC ICs. This is because the goal of the Flagship run was to ensure that each group can handle the MUSIC ICs properly, *not* to squeeze out the first paper in a hasty way. So far, our idea has paid off well in that all the participating codes have worked out how to read in the common ICs, and 6(7) out of 8 code groups have successfully performed RUN-1 (~300 pc resolution run; see description **here**) in a reasonable amount of computing time. After more groups complete their RUN-1 simulation, we will soon collectively move to RUN-2 (~100 pc resolution run). **< ACTION ITEM 1 > The Steering Committee asks the participants of the proof-of-concept Flagship run to please start RUN-2 (~100 pc resolution run), and report the progress in the Workspace (collaborative doc).**

		– To produce simple figures and profiles for the Flagship paper (see Section 5.3 of the paper linked above), we will also adopt a philosophy of letting each group to take as much time as needed to figure out how to use yt as its preferred choice of analysis. This is again because the goal of the Flagship run is to build up and field-test the pipeline of the Project, including the common analysis to be used in the future science-oriented projects, *not* to quickly produce a single paper. If each group could successfully work out how to read in and analyze the dataset using **yt toolkit** in the Flagship run, this would greatly expedite the analysis required in the subsequent science-oriented papers. We are very optimistic about this prospect as more codes are now able to utilize yt, thanks in part to a venue like the recent **yt workshop**. See the progress report of WG IV below. **< ACTION ITEM 2 >** **Therefore, the Steering Committee also asks each code group, spearheaded by the participants of the Flagship run, to please make all-out efforts to utilize yt in the analysis of the Flagship paper.**
II	Common ICs: Isolated	*[1] Isolated Disk ICs* – Please see Section 2.2 of the paper linked above for detailed information.
III	Common ICs: Cosmological	*[1] Cosmological ICs* – Please see Section 2.1 of the paper linked above for general description. Please also see **WG III's page** for detailed information on all eight selected halos. – A new IC for a ~1e11 Ms halo based on the Planck cosmology is currently being tested (h=0.678, $\mathrm{Omega_M}$=0.308, $\mathrm{Omega_{Lambda}}$=0.692, $\mathrm{Omega_b}$=0.0482, n_s=0.961 and σ_8=0.826; see **this table** or Fig.5 of **this paper**). One may find very preliminary results of this test in **WG III's page**. *[2] Companion Paper to Describe ICs* – In order to supplement the Flagship paper, WG II and III are (jointly) preparing a companion paper(s), which describes the initial conditions in detail so anybody can download the ICs and perform their own simulations regardless of their affiliation with the Project. Because the Flagship paper will provide only the basic analysis of the DM-only proof-of-concept Flagship run of a ~1e11 Ms halo, more details of its result will also be presented in the companion paper including the comparison of the halo catalogues, dark matter merger histories, power spectrum, etc. See Section 5.3 of the paper linked above. *[3] MUSIC Support for Minimum Volume High-resolution Region* – The highest-resolution region of the MUSIC IC no longer needs to be in a cuboidal shape. Instead, MUSIC now supports an ellipsoidal shape for selected codes. Please see Section 3-C of **this page** (proof-of-concept Flagship run) for more discussion.
IV	Common Analysis	*[1] Guiding Principles for Common Analysis* – Please see Section 4.2 of the paper linked above. *[2] Code Support in yt Analysis Toolkit* – All in all **yt toolkit** has varying degrees of support for the following types of data: ARTIO, ART, RAMSES, Enzo, Gadget, Gasoline/Pkdgrav, Nyx. – There are several steps of getting these supported at the fundamental level: 1) Reading the data and spatially selecting. Our understanding is that this currently works for **all** the codes including Tipsy, Gadget, and Gadget-hdf5. But there may be some caveats for the particle data. 2) Making images works for **all** AMR codes including ARTIO, ART, and RAMSES (i.e., not just patch-based) thanks to the active participation by ART-I/II developers. Images **can** be made from the data of the particle-based codes, but not nicely. 3) Units: This is where WG IV will need the most help. In general, we will need volunteers from the SPH codes in

		WG IV who can represent their codes with intimate knowledge of the inner workings (e.g., how to deal with the data formats). - The next step is going to be emphasizing and implementing YTEP-0013, which will be the basis for implementing particle-based codes in yt. This will enable a consistent internal API for constructing fluid fields based on particles.
V	Isolated Galaxies and Subgrid Physics	*[1] Testing Subgrid Implementations in Isolated Disk ICs* - Please see Section 2.2 and Appendix A.1 of the paper linked above. Please also see the collaborative doc for more information. - Subgrid prescriptions for star formation and feedback in various code platforms will be tuned using an isolated disk simulation so it provides a realistic interstellar and circumgalactic medium. (This is why Section 3 of the paper does not discuss star formation and feedback.)

2. Resolution for DM-only Proof-of-Concept Flagship Run

- Do we agree on the choice of force softening/resolution for the DM-only proof-of-concept Flagship Runs?

- [Proposed Resolution for RUN-2]
  >> For particle codes: gravitational softening length of 1.14 comoving kpc until z=9, 114 physical pc afterwards
  >> For AMR codes: finest cell size of 163 comoving pc all the way to z=0 (i.e. maximum refinement level = 12)
  >> See Fig.B1 of the Aquila paper for comparison.

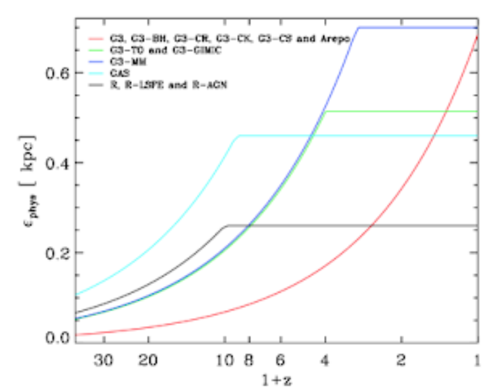

Figure B1. Evolution of the *physical* gravitational softening in the different models.

3. Strategy to Handle the Disagreements During Comparisons

- How do we want to treat the disagreements between the codes during the AGORA projects? Will the submitted results be permitted to be fixed? (Raised by Nick Gnedin)

- [Proposed Policy]
  The goal of the project is to compare *correct* results. Therefore, if any problems/mistakes were discovered and the participant wishes, he/she

should be given a chance to go back and fix their codes/bugs and resubmit the result. While this policy may mean a delay of publication in some cases, it will ensure not only that a comparison is made in a fair and scientifically-meaningful way, but also that a smaller code group can feel safe when they first submit their simulation results. (These small code groups may not necessarily have many developers to double-check the results before the submission!)

4. Strategy for Data Storage

- Which data, how often, and where do we want to store the snapshots of the performed simulations? Should we already start building a central repository for sharing the data from the proof-of-concept Flagship Runs?

- [Proposed Policy]
  Please see Section 4.1 of the paper linked above.

- Each of the participating codes will generate large quantities of unprocessed, intermediate data, in the form of "checkpoints" describing the state of the simulation at a given time (200 timesteps equally spaced in expansion parameter + redshift snapshots at z = 6, 3, 2, 1, 0.5, 0.2, 0.0 at the very least). Each group is also advised to store additional outputs at slightly earlier redshift, shifted by $\Delta z \sim 0.05$. This extra information will help us investigate whether an inter-platform offset in time-stepping causes timing differences for halo mergers, for example. **--> Outdated. See the new strategy here.**

- Storage and post-processing compute time is planned to be available at the San Diego Supercomputer Center at UCSD, the Hyades system at UCSC, and/or the Data-Scope system at John Hopkins. An announcement will follow soon as official agreements are made between the institutions.

5. Strategy to Compare the Computational Efficiency

- Do we plan to compare the computational efficiency of the codes at some point? Should we keep the records of the computational cost of participating codes? While it is definitely not in the scope of the Flagship paper, some participants might be interested in ballpark runtimes of other codes.

- [Proposed Policy]
  We will not discuss the computational efficiency in AGORA publications in the foreseeable future. However, when asked, participants are encouraged, but not required, to share the runtimes of your code within the collaboration.

6. New Website

- A new public front of the AGORA Project, **www.AGORAsimulations.org**, is purchased and now up and running. We thank UC-HiPACC for financial support, and Nina McCurdy and Ramon Berger for design and technical support. While this address at the moment only redirects visitors to two pre-existing AGORA webpages on Google, it will soon be updated so it allows visitors to access the AGORA simulations data repository.

7. Next Workshop

- Preliminary dates:
  >> The annual Santa Cruz Galaxy Workshop 2013 (Aug. 12-16, tentatively)
  >> The 2nd AGORA Workshop 2013 (Aug. 16-18, tentatively; 3-day weekend meeting just like **last year**)